

When ML Meets Market Reality

Stacking SATS: How a 296K-Parameter CNN Got Beat by a Coin Flip

GT OMSA Practicum · Trilemma Foundation · Bitcoin Accumulation Tournament

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The Tournament: Daily Bitcoin Allocation



OBJECTIVE

Predict daily BTC allocation weights
across 2016–2025 (~3,300 trading days)



METRIC

Recency-Weighted SPD Percentile
Target: top 60% of all submissions



CONSTRAINTS

$\sum w_i = 1.0$ · $w_i \geq 1e-5$

No shorting · Full budget each day



CRITICAL RULE

Zero lookahead — strict causality
Features use only data through day t

Why Visual Trading Systems?



FINANCE ROOTS

MBA-era chart reading
(Magee & Edwards)

Human traders 'see' patterns
that quants miss



RESEARCH SIGNAL

Industry reports show
high F1 on candlestick
classification tasks

Explored in OMSA with
Prof. Baich



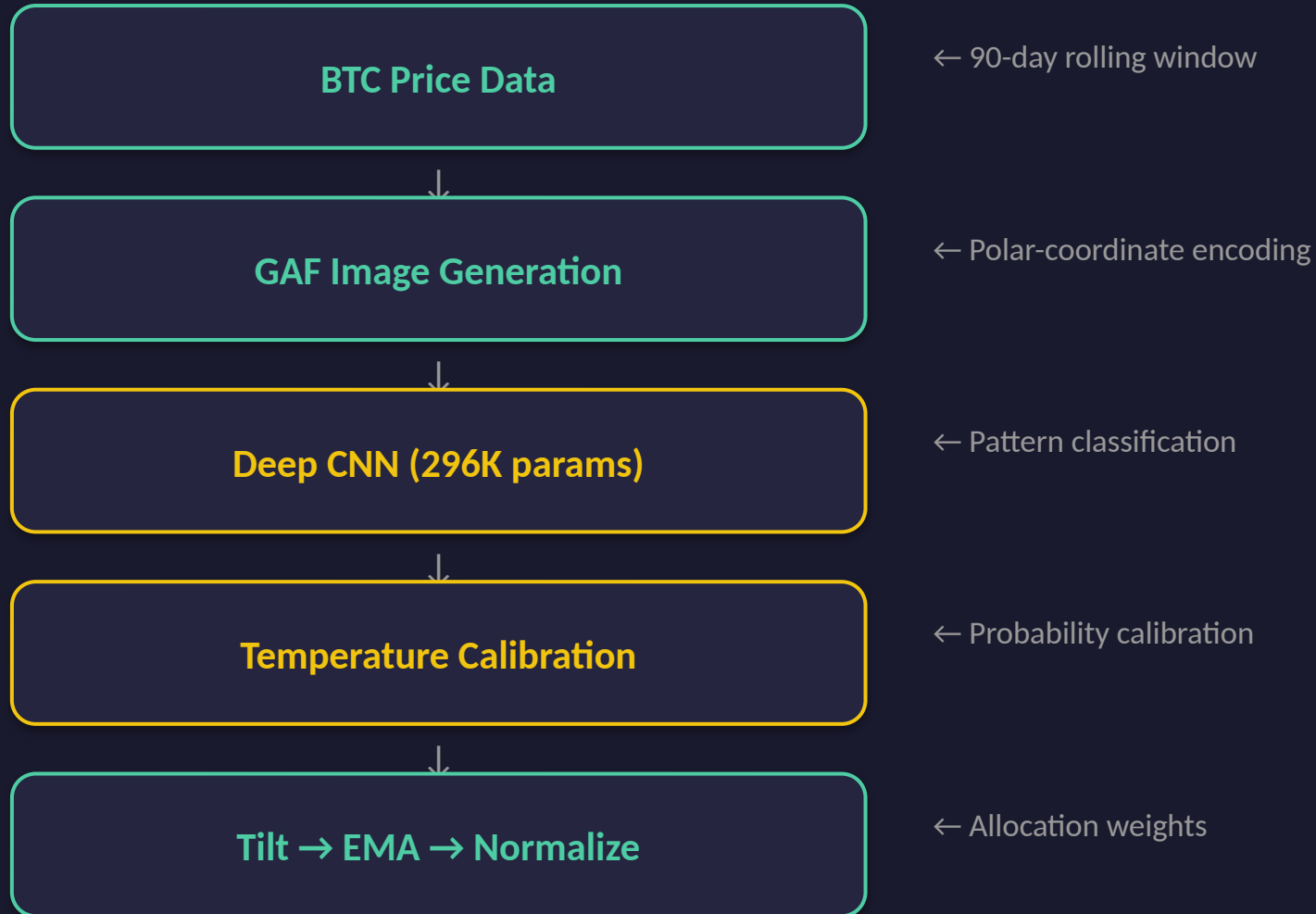
THE HYPOTHESIS

Pattern recognition $\neq$
Profitable trading

Does academic success
survive strict causality?

Hypothesis: GAF images encode price patterns that a CNN can exploit for superior BTC accumulation under strict causality constraints

Our Initial Approach: The Pipeline



OUTPUT: Normalized daily weights · $\sum w = 1.0$ · seed = 42 · Zero lookahead

Reality Check: First Results

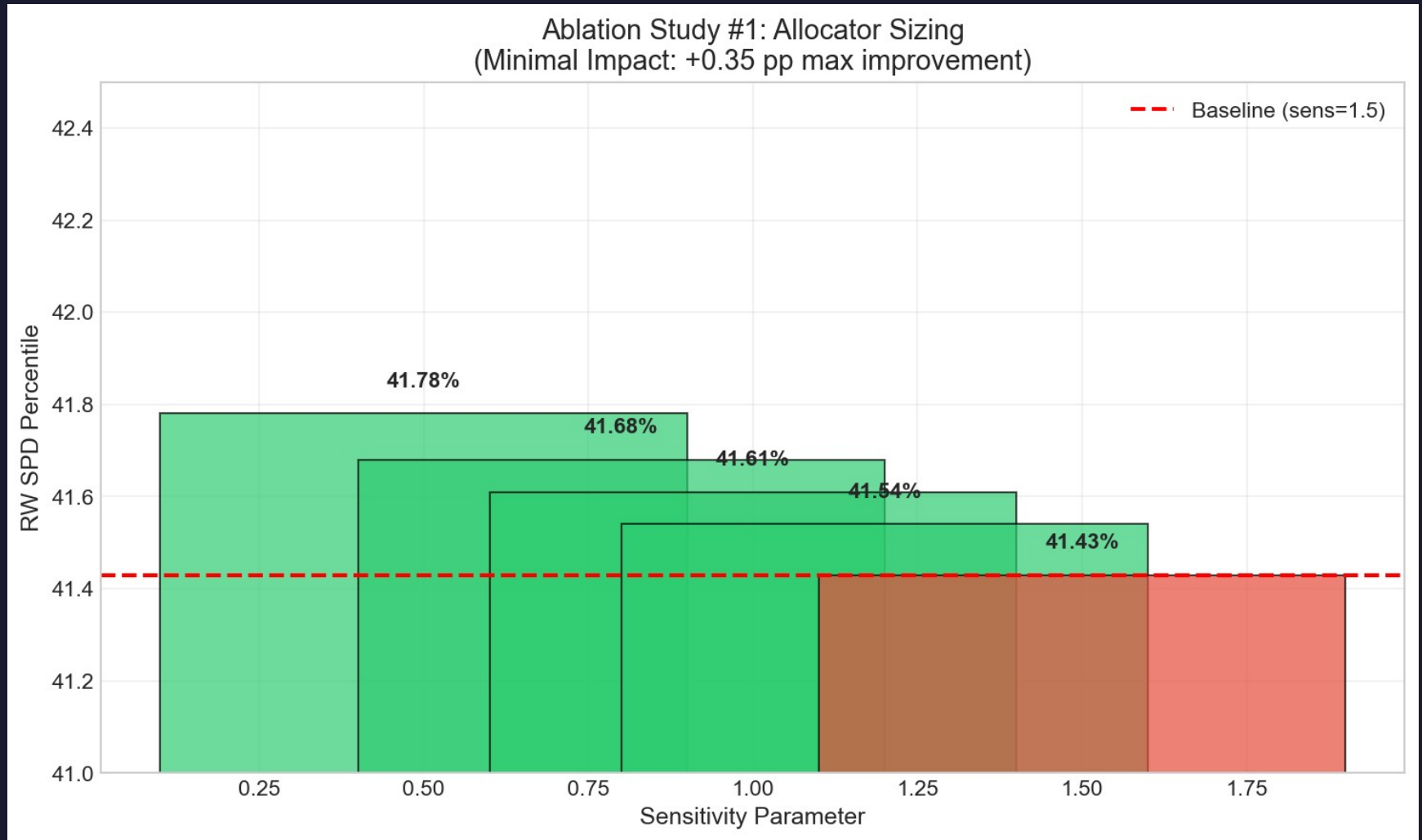
The 296K-parameter CNN barely outperforms a constant neutral signal — and both fall well short of the 60% target

Metric	CNN (296K)	Simplified	Target
RW SPD Percentile	41.43%	41.94%	> 60%
Win Rate vs DCA	54.32%	70.42%	> 50%
Execution Time	~17 min	~2.5 min	—
Model Size	1.1 MB	None	—
Reproducibility	seed=42	seed=42	✓

 Neutral baseline = constant prob_{up} = 0.5 → allocator pipeline unchanged; no model involved

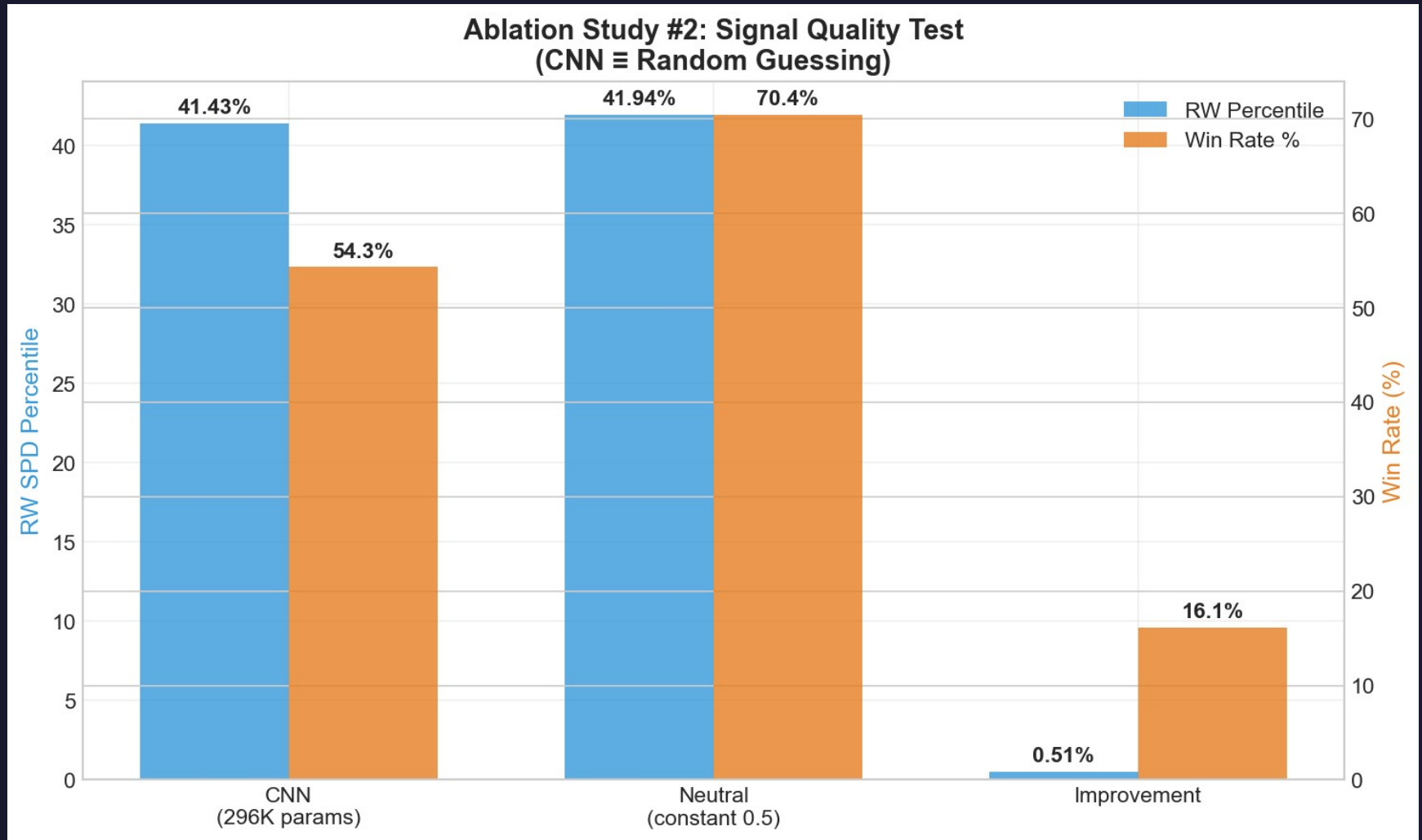
 RW-SPD rewards when and how much you deviate from neutral — not just direction

Ablation #1 — Allocator Sensitivity



Verdict: Sweeping allocator aggressiveness moves RW percentile < 0.4 pp → NOT the root cause

Ablation #2 — The "Aha!" Moment



🚫 CNN signal $\approx$ neutral baseline $\rightarrow$ no predictive edge detected

✅ Neutral: +16 pp win rate · simpler · 7× faster

Midterm Diagnosis: Why GAF + CNN Failed



TIME HORIZON MISMATCH

GAF lookback: 90 days
BTC cycles: 6–18 months
Misses bull/bear regime shifts



DAILY NOISE

Daily prices = low SNR
Tournament forces daily decisions
Amplifies noise, destroys signal



WRONG ARCHITECTURE

CNN exploits spatial patterns
Finance needs temporal modeling
Translation invariance irrelevant



TRAINING MISMATCH

Train window: 2014–2015 (available historical)
Test: 2016–2025 (tournament span)

Out-of-distribution: regime shift post-2016

What We Learned (Technical)



DAILY SNR IS LOW

Daily price moves are dominated by noise, not signal

Any model exploiting day-to-day patterns likely fits noise.
Weekly or regime-horizon signals have better SNR.



WRONG ARCHITECTURE

GAF + CNN exploits spatial patterns via 2D convolutions

BTC accumulation needs temporal and regime modeling, not translation invariance.
Problem is sequential, not visual.



ALLOCATOR CAN'T RESCUE WEAK SIGNAL

Sweeping tilt & EMA params gave < 0.5 pp improvement

Garbage in → garbage out: allocator amplifies signal, it doesn't create it.
Test signal quality first.

Implication: Test 'does this signal beat random?' before optimizing any downstream component

The Evidence-Based Pivot

CNN (296K params)

RW Percentile

41.43%

Win Rate

54.32%

Exec Time

~17 min

Model Size

1.1 MB

VS

Simplified (neutral)

RW Percentile

41.94%

Win Rate

70.42%

Exec Time

~2.5 min

Model Size

None



Simplified wins on ALL dimensions

What We Did Right

✓ STRICT CAUSALITY

Last-row test · Purge test
Shift test — all PASSED
Zero lookahead confirmed

✓ DETERMINISTIC

seed = 42
Fully reproducible output
Grader-safe implementation

✓ SCIENTIFIC RIGOR

Hypothesis → test → conclude
Systematic ablation studies
Transparent negative results

✓ 62/62 TESTS PASSING

Unit + integration + ablation
Budget accounting tests
All results documented

Roadmap Forward

MIDTERM

Status: DONE

- Neutral baseline
- 62/62 tests passing
- All ablations complete
- Causality verified

PHASE I

→ March 2026

- Regime-gated allocator
- (HMM / volatility regime)
- Weekly signal horizon
- XGBoost baseline

PHASE II

→ Final Submission

- Ensemble integration
- Walk-forward backtest
- Final report + model card
- Full documentation

Goal: attempt to exceed 60%+ RW percentile via regime gating + weekly signals (to be validated in Phase I)

THE BOTTOM LINE

"We built a technically sophisticated system and discovered through rigorous testing that it performs statistically indistinguishable from random under the RW-SPD tournament metric."

In financial time series, signal quality matters more than model complexity."

COMPLEXITY $\neq$ BETTER

296K params matched
by a constant signal

PROCESS IS THE VALUE

Scientific integrity
even with null results

SIGNAL FIRST

Test 'beats random?'
before optimizing

Questions & Discussion

Q1 • Metric Interpretation

Is our understanding of RW-SPD percentile vs win rate correct?

Q2 • Dataset Shift

How would you recommend handling regime shifts / non-stationarity under strict causality?

Q3 • Pivot Scope

Is weekly signals + regime detection appropriately scoped for Phase I?

Q4 • Baseline Choice

Keep simplified neutral baseline, or invest in XGBoost / ensemble?